

# Introduction to Retrieval-Augmented Generation

*A short guide to building RAG applications*

## 1. What is RAG?

Retrieval-Augmented Generation (RAG) combines a retrieval system with a large language model. Instead of relying only on what the model memorized during training, the system first searches a knowledge base for relevant passages and then passes them to the model as context. This reduces hallucinations and lets the model answer questions about private or recent data.

## 2. The Ingestion Pipeline

Ingestion prepares documents for search. The usual steps are: load the document, extract text, clean it, split it into chunks, create embeddings for each chunk, and store the vectors in a vector database. Chunk size and overlap strongly affect retrieval quality.

### 3. Chunking Strategies

Common strategies include fixed-size chunking, recursive character splitting, token-based splitting, structure-aware splitting (by headings), and semantic chunking. Recursive splitting is a good default because it tries to keep paragraphs and sentences intact.

| Strategy            | Best for            | Typical chunk size |
|---------------------|---------------------|--------------------|
| Recursive character | General text        | 500-1000 chars     |
| Token-based         | Strict LLM limits   | 256-512 tokens     |
| Header-based        | Markdown / HTML     | Per section        |
| Semantic            | Topic-shifting docs | Variable           |

### 4. Embeddings and Vector Stores

An embedding model converts each chunk into a numeric vector that captures its meaning. Vector stores such as FAISS, Chroma, Pinecone, and pgvector index these vectors so that the most similar chunks to a query can be found quickly using cosine similarity or dot product.

## 5. Evaluation

RAG quality is measured on two sides. Retrieval metrics include recall at k and mean reciprocal rank. Generation metrics include faithfulness, answer relevance, and context precision. Always evaluate with a fixed set of test questions so changes to chunking or prompts can be compared fairly.

## 6. Common Pitfalls

Frequent problems are chunks that are too large or too small, missing metadata, scanned PDFs with no text layer, tables flattened into unreadable text, and ignoring page numbers needed for citations.